2026 VCE Specialist Mathematics 1 (NHT) external assessment report

This report provides sample answers, or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

Question 1a.

Using $s = \frac{u+v}{2}t$:

$$10 = \frac{u+7}{2} \times 2$$
$$= u + 7$$

Therefore, $u = 3$.

Question 1b.

Find the acceleration of the object. Using $v = u + at$:

$$7 = 3 + 2a$$
$$\Rightarrow 4 = 2a$$
$$\Rightarrow a = 2$$

Using $s = ut + \frac{1}{2}at^2$:

$$28 = 3t + \frac{1}{2} \times 2t^2$$

Then:

$$t^2 + 3t - 28 = 0$$
$$(t+7)(t-4) = 0$$

Therefore, $t = 4$ (since $t \geq 0$).

Question 2

Let $P(n)$ be the proposition that n^2+5n is even for all $n \in N$.

$P(1): 1+5=6$ is even and so $P(1)$ is true.

Assume $P(k)$ is true for some $k \in N$. That is: $k^2+5k=2m$ for some $m \in N$.

$P(k+1)$:

$$\begin{aligned}(k+1)^2+5(k+1) &= k^2+2k+1+5k+5 \\ &= \underbrace{k^2+5k}_{=2m}+2k+6 \\ &= 2m+2(k+3) \\ &= 2(m+k+3)\end{aligned}$$

which is even. So $P(k)$ is true implies that $P(k+1)$ is true.

Therefore, by the principle of mathematical induction, $P(n)$ is true for all $n \in N$.

Question 3

$$f'(x)=5x^4+4x^3-1$$
$$f''(x)=20x^3+12x^2=4x^2(5x+3)$$

$f''(x)=0 \Rightarrow x=0$ or $x=-\frac{3}{5}$. A table of values can be used to establish the nature of the points:

x	-1	$-\frac{3}{5}$	$-\frac{1}{5}$	0	1
$f''(x)$	$-8<0$	0	$\frac{8}{25}>0$	0	$32>0$

There is a change in concavity at $x=-\frac{3}{5}$. There is no change in concavity at $x=0$.

Therefore the x-coordinate of the point of inflection is $x=-\frac{3}{5}$.

Question 4a.

Substituting $y = e^{mx}$ into the differential equation gives:

$$\begin{aligned} & 2m^2e^{mx} - me^{mx} - e^{mx} = 0 \\ & \quad \Rightarrow 2m^2 - m - 1 = 0 \quad \left(e^{mx} \neq 0\right) \\ & \Rightarrow (2m+1)(m-1) = 0 \end{aligned}$$

giving $m = -\frac{1}{2}$ or $m = 1$.

Question 4b.

The volume V is

$$\begin{aligned} \pi \int_0^1 \left(\left(e^x\right)^2 - \left(e^{-\frac{1}{2}x}\right)^2 \right) dx &= \pi \int_0^1 \left(e^{2x} - e^{-x}\right) dx \\ &= \pi \left[\frac{1}{2}e^{2x} + e^{-x} \right]_0^1 \\ &= \pi \left(\frac{1}{2}e^2 + e^{-1} \right) - \pi \left(\frac{1}{2} + 1 \right) \\ &= \pi \left(\frac{1}{2}e^2 + e^{-1} - \frac{3}{2} \right) \\ &= \frac{\pi}{2}\left(e^2 + 2e^{-1} - 3\right) \end{aligned}$$

Question 5

Let $u = x+1$. Then, $du = dx$ and $u^2 = x^2 + 2x + 1$.

When $x = 0$, $u = 1$ and when $x = 3$, $u = 4$.

The integral becomes

$$\begin{aligned}\int_1^4 \frac{u^2-1}{\sqrt{u}}du &= \int_1^4 \left(u^{\frac{3}{2}} - u^{-\frac{1}{2}}\right)du \\ &= \left[\frac{2}{5}u^{\frac{5}{2}} - 2u^{\frac{1}{2}}\right]_1^4 \\ &= \left(\frac{2}{5}\times 2^5 - 4\right) - \left(\frac{2}{5} - 2\right) \\ &= \frac{64}{5} - \frac{20}{5} - \frac{2}{5} + \frac{10}{5} \\ &= \frac{52}{5}\end{aligned}$$

Question 6a.

The sample mean is:

$$\bar{x} = \frac{11.51+18.09}{2} = \frac{29.60}{2} = 14.8$$

Question 6b.

Using (for example) the right-hand side of the confidence interval:

$$\begin{aligned}14.8 + 1.645\times\frac{6}{\sqrt{n}} &= 18.09 \\ \Rightarrow 1.645\times\frac{6}{\sqrt{n}} &= 18.09 - 14.08 = 3.29 \\ \Rightarrow \frac{6}{\sqrt{n}} &= 2 \\ \Rightarrow 3 &= \sqrt{n}\end{aligned}$$

The sample size is $n = 9$.

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Question 7a.

$\frac{dx}{dt} = 2t - \frac{2}{t^3}$ and $\frac{dy}{dt} = 2t + \frac{2}{t^3}$. So, by the chain rule:

$$\frac{dy}{dx} = \frac{2t + \frac{2}{t^3}}{2t - \frac{2}{t^3}} \times \frac{t^3/2}{t^3/2}$$
$$= \frac{t^4 + 1}{t^4 - 1}$$

The gradient is undefined when $t = \pm 1$. In both cases, the coordinates of the point on the curve where the tangent is vertical is $(2, 0)$.

Question 7b.

$$x^2 - y^2 = \left(t^2 + \frac{1}{t^2}\right)^2 - \left(t^2 - \frac{1}{t^2}\right)^2$$
$$= t^4 + 2 + \frac{1}{t^4} - t^4 + 2 - \frac{1}{t^4}$$
$$= 4$$

So, the Cartesian equation of the curve is $x^2 - y^2 = 4$.

Question 8

Use integration by parts:

$$u = \log_e(2x) \qquad \frac{dv}{dx} = 4x^3$$
$$\frac{du}{dx} = \frac{1}{x} \qquad v = x^4$$

Then:

$$f(x) = x^4 \log_e(2x) - \int x^3 \, dx$$
$$= x^4 \log_e(2x) - \frac{1}{4}x^4 + c$$

Then $f\left(\frac{1}{2}\right) = -\frac{1}{4} \times \frac{1}{16} + c = 0 \Rightarrow c = \frac{1}{64}$. So:

$$f(x) = x^4 \log_e(2x) - \frac{1}{4}x^4 + \frac{1}{64}$$

Question 9a.

Using (for example) polynomial long division:

$$
\begin{array}{r}
x^2-\frac{1}{2}x-2 \\
2x-2\overline{)\,2x^3-3x^2-3x+2} \\
\underline{2x^3-2x^2} \\
-x^2-3x \\
\underline{-x^2+x} \\
-4x+2 \\
\underline{-4x+4} \\
-2
\end{array}
$$

Therefore, $f(x)=x^2-\frac{1}{2}x-2-\frac{2}{2x-2}$.

The equation of the parabolic asymptote is $y=x^2-\frac{1}{2}x-2$.

Question 9b.

$x=-1$ is a solution of $2x^3-3x^2-3x+2=0$. Factorise:

$$2x^3-3x^2-3x+2=(x+1)\left(2x^2+\alpha x+2\right)$$

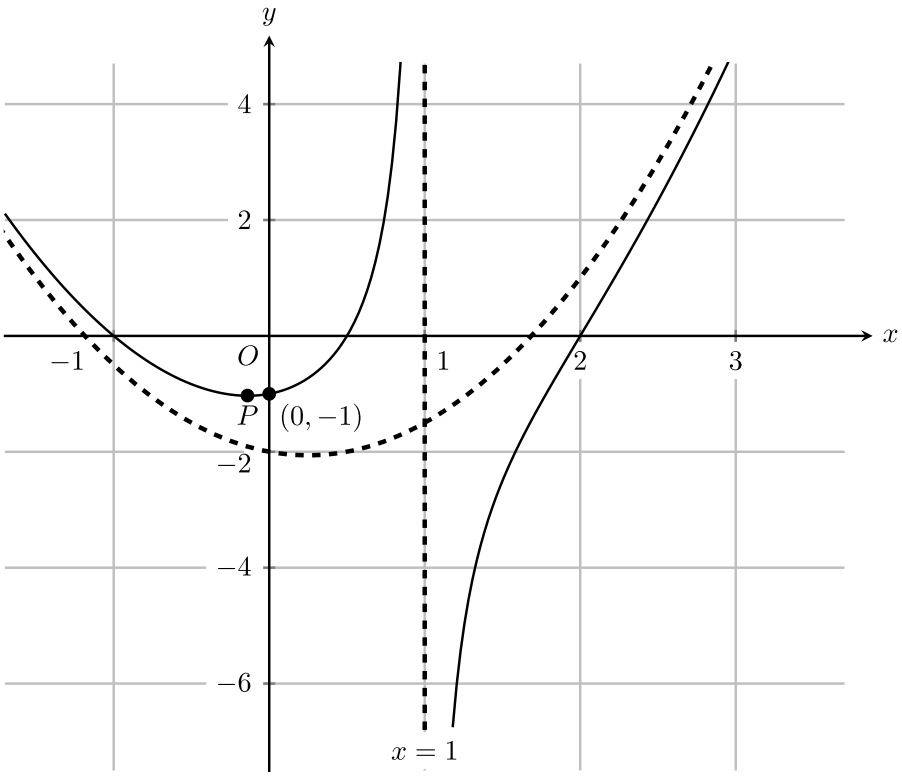
Compare coefficients to find $\alpha=-5$. Therefore:

$$(x+1)\left(2x^2-5x+2\right)=(x+1)(2x-1)(x-2)=0$$

and so, the x-intercepts of the graph occur at $x=-1$, $x=\frac{1}{2}$ and $x=2$.

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Question 9c.



Question 10

Since the coefficients of $P(z)$ are real, the conjugate root theorem applies and so $P(-i)=0$.

A quadratic factor of $P(z)$ is $(z-i)(z+i)=z^2+1$.

As there are three distinct solutions, the remaining solution must be real and must appear in the form of a repeated root:

$$P(z)=\left(z^2+1\right)(z-a)^2$$

By inspection, $a=\pm 1$.

If $a=1$:

$$\begin{aligned}P(z)&=\left(z^2+1\right)(z-1)^2\\&=\left(z^2+1\right)\left(z^2-2z+1\right)\\&=z^4-2z^3+2z^2-2z+1\end{aligned}$$

giving $b=-2$, $c=2$ and $d=-2$.

If $a=-1$:

$$\begin{aligned}P(z)&=\left(z^2+1\right)(z+1)^2\\&=\left(z^2+1\right)\left(z^2+2z+1\right)\\&=z^4+2z^3+2z^2+2z+1\end{aligned}$$

giving $b=2$, $c=2$ and $d=2$.

Question 11

Let $\underset{\sim}{n}_1 = 2\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k}$ and $\underset{\sim}{n}_2 = p\underset{\sim}{i} + (p+1)\underset{\sim}{j} + 2\underset{\sim}{k}$. Then:

$$\underset{\sim}{n}_1 \cdot \underset{\sim}{n}_2 = 2p + p + 1 + 2 = 3p + 3$$
$$|\underset{\sim}{n}_1| = \sqrt{6}$$
$$|\underset{\sim}{n}_2| = \sqrt{2p^2 + 2p + 5}$$

The angle is 60° and so:

$$\frac{3p+3}{\sqrt{6}\sqrt{2p^2+2p+5}} = \cos\left(60^\circ\right) = \frac{1}{2}$$
$$\Rightarrow 6(p+1) = \sqrt{6}\sqrt{2p^2+2p+5}$$
$$\Rightarrow 6\left(p^2+2p+1\right) = 2p^2+2p+5$$
$$\Rightarrow 4p^2 + 10p + 1 = 0$$

Therefore:

$$p = \frac{-10 \pm \sqrt{100-16}}{8}$$
$$= \frac{-10 \pm \sqrt{84}}{8}$$
$$= \frac{-5 \pm \sqrt{21}}{4}$$